

Updated: May 2026

Habeeb Olufowobi, PhD

Assistant Professor

📍 Department of Computer Science and Engineering
The University of Texas at Arlington
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👤 Professional Summary

Habeeb Olufowobi is an Assistant Professor in the Department of Computer Science and Engineering at The University of Texas at Arlington, where he directs the Cyber-physical Systems Security Lab (CSS Lab). His research focuses on the security, resilience, and trustworthiness of safety-critical cyber-physical and intelligent systems, with emphasis on autonomous vehicles, embedded and real-time systems, Internet of Things (IoT) infrastructures, and AI-enabled edge intelligence. His work investigates how adversarial, real-time, and cross-layer threats impact the interaction between sensing, communication, learning, and control systems, and develops principled defenses that improve the safety, resilience, and trustworthiness of next-generation intelligent systems.

📁 Current Appointment

Assistant Professor	August 2020 – Present
Department of Computer Science and Engineering The University of Texas at Arlington, TX	

📁 Past Appointments

Lecturer	August 2019 – May 2020
Department of Electrical Engineering and Computer Science Howard University, Washington, DC	
Graduate Research Assistant / Project Lead	January 2016 – August 2020
Embedded Systems Security Lab Howard University, Washington, DC	
Research Intern	June 2017 – August 2017
Oak Ridge National Laboratory	
Research Intern	June 2016 – August 2016
IBM Almaden Research	
Project Management Intern	June 2014 – September 2014
Synaptics Inc.	

Education

Ph.D. in Computer Science

May 2019

Howard University, Washington, DC

Concentration: Embedded Systems Security

Dissertation: *Fail-Operational Intrusion Detection Systems (FO-IDS): A Mechanism for Securing Automotive In-Vehicle Networks*

M.S. in Electrical Engineering

December 2014

California State Polytechnic University, Pomona

Concentration: Computer Systems

B.S. in Computer Science (with honors)

July 2011

Fountain University, Osogbo, Nigeria

Research Interests – CPS Security and Trustworthy AI

- Security and Resilience of Autonomous and Connected Vehicle Systems
- Embedded, Real-Time, and Cyber-Physical Systems (CPS) Security
- Trustworthy Artificial Intelligence, Explainability, and Adversarial Machine Learning
- Intelligent Edge Systems, IoT Security, and Networked Infrastructures
- Secure and Explainable AI for Healthcare and Safety-Critical Applications
- Privacy, Fairness, and Bias Mitigation in AI/ML Systems

My research focuses on improving the security, resilience, and trustworthiness of cyber-physical and AI-enabled systems operating in safety-critical environments. My work spans multiple application domains, including connected and autonomous vehicles, embedded and real-time systems, intelligent transportation systems, distributed and edge computing, and healthcare AI. I investigate vulnerabilities and defense mechanisms across the hardware–software stack, with emphasis on adversarial, real-time, and cross-layer threats affecting autonomous and intelligent systems.

In addition, my research explores interpretability, explainability, fairness, and bias mitigation in AI/ML models used in healthcare and other high-impact domains. I am particularly interested in developing secure and trustworthy intelligent systems that operate reliably under adversarial conditions while maintaining transparency, robustness, and privacy guarantees. My research has been supported by the National Science Foundation (including the NSF CAREER Award), the National Institutes of Health AIM-AHEAD initiative, and the University of Texas System.

Teaching Interests

I develop and teach practical, hands-on, and research-oriented courses in cybersecurity, computer and network security, cloud computing, embedded and real-time systems, cyber-physical systems security, IoT systems, and trustworthy AI. My teaching emphasizes experiential learning, applied problem solving, and the integration of real-world security challenges into classroom instruction, laboratory environments, and research-driven projects.

Grants Awarded

National Science Foundation (NSF)	\$598,895
<i>CAREER: Securing Connected Autonomous Vehicle Stack Against Adversarial Input</i>	
Principal Investigator	07/2025–06/2030
NIH AIM-AHEAD	\$250,000
<i>Data-Driven Diagnosis and Prediction of Heart Disease Outcomes in Chagas Disease with Ethical and Explainable AI/ML Models</i>	
Principal Investigator	10/2024–03/2026
Google-CAHSI	\$25,000
<i>Google-CAHSI Cybersecurity Curriculum Fellowship</i>	
Principal Investigator	10/2024–05/2026
NIH AIM-AHEAD	\$50,000
<i>Improving Health Equity in Algorithmic Decision-Making Using Explainable AI Techniques</i>	
Principal Investigator	09/2023–09/2024

Publications

Note: *Underlined names indicate lead student authors I advise or co-advise.*

Journals and Magazines

- J10. **OJ-COMS'26** Md Salik Parwez, Sai Teja Srivillibhutturu, Abdullah Al-Shammari, and **Habeeb Olufowobi**, “Digital Twin–Guided AI Path Planning for Connectivity-Aware Mobility,” *IEEE Open Journal of the Communications Society*, 2026.
- J9. **ACCESS'24** Christopher Kolimago, Michael Robson, **Habeeb Olufowobi**, and Ebelechukwu Nwafor, “Simulating Load Sharing for Resource Constrained Devices,” *IEEE Access*, 2024.
- J8. **T-ITS'24** Paul Agbaje, **Habeeb Olufowobi**, Sena Hounsinnou, and Gedare Bloom, “From Weeping to Wailing: A Transitive Stealthy Bus-Off Attack,” *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- J7. **IoTJ'24** Afia Anjum, Paul Agbaje, Sena Hounsinnou, Nadra Guizani, and **Habeeb Olufowobi**, “D-NDNoT: Deterministic Named Data Networking for Time-Sensitive IoT Applications,” *IEEE Internet of Things Journal*, 2024.
- J6. **FGCS'23** Afia Anjum, Paul Agbaje, Arkajyoti Mitra, Emmanuel Oseghale, Ebelechukwu Nwafor, and **Habeeb Olufowobi**, “Towards Named Data Networking Technology: Emerging Applications, Use Cases, and Challenges for Secure Data Communication,” *Future Generation Computer Systems*, 2023.
- J5. **TVT'23** Omolade Ikumapayi, **Habeeb Olufowobi**, Jeremy Daily, Tingting Hu, Ivan Cibrario Bertolotti, and Gedare Bloom, “CANASTA: Controller Area Network Authentication Scheduling Timing Analysis,” *IEEE Transactions on Vehicular Technology*, 2023.

- J4. **T-ITS'22** Paul Agbaje, Afia Anjum, Arkajyoti Mitra, Emmanuel Oseghale, Gedare Bloom, and **Habeeb Olufowobi**, “Survey of Interoperability Challenges in the Internet of Vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, 2022.
- J3. **CEM'20** Uchenna Ezeobi, **Habeeb Olufowobi**, Clinton Young, Joseph Zambreno, and Gedare Bloom, “Reverse Engineering Controller Area Network Messages using Unsupervised Machine Learning,” *IEEE Consumer Electronics Magazine*, 2020.
- J2. **TVT'19** **Habeeb Olufowobi**, Clinton Young, Joseph Zambreno, and Gedare Bloom, “SAID-uCANT: Specification-based Automotive Intrusion Detection using Controller Area Network Timing,” *IEEE Transactions on Vehicular Technology*, 2019.
- J1. **D&T'19** Clinton Young, **Habeeb Olufowobi**, Joseph Zambreno, and Gedare Bloom, “Survey of Automotive Controller Area Network Intrusion Detection Systems,” *IEEE Design & Test*, 2019.

✔ Refereed Conference Articles

- C30. **ICML'26** Paul Agbaje and **Habeeb Olufowobi**, “Revisiting Asymmetries in Black-box Link Stealing against Graph Neural Networks,” *International Conference on Machine Learning (ICML)*, PMLR, 2026.
- C29. **PETS'26** Paul Agbaje, Afia Anjum, Arkajyoti Mitra, and **Habeeb Olufowobi**, “Unveiling Graph Copycats: Inference Attacks with Student Models,” *Proceedings on Privacy Enhancing Technologies*, 2026.
- C28. **ITSC'26** Md Salik Parwez, Sai Teja Srivillibhutturu, Kushal Shah, and **Habeeb Olufowobi**, “Efficiency–Equity Trade-Off Characterization in Hierarchical Traffic Congestion Pricing,” *IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, 2026.
- C27. **ICC'26** Md Salik Parwez, Sai Teja Srivillibhutturu, Sai Venkat Reddy Kopparthi, Asfiya Misba, Debashri Roy, **Habeeb Olufowobi**, and Charles Kim, “CTMap: LLM-Enabled Connectivity-Aware Path Planning in Millimeter-Wave Digital Twin Networks,” *IEEE International Conference on Communications (ICC)*, 2026.
- C26. **IV'26** Pedram MohajerAnsari, Amir Salarpour, Michael Kühr, Siyu Huang, Mohammad Hamad, Sebastian Steinhorst, **Habeeb Olufowobi**, and Mert D. Pesé, “Toward Inherently Robust VLMs Against Visual Perception Attacks,” *IEEE Intelligent Vehicles Symposium*, 2026.
- C25. **WACV'26** Aqsa Yousaf, Sint Sint Win, Megan Coffee, and **Habeeb Olufowobi**, “MorphXAI: An Explainable Framework for Morphological Analysis of Parasites in Blood Smear Images,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.
- C24. **3DV'26** Aqsa Yousaf, Paul Agbaje, Afia Anjum, Arkajyoti Mitra, and **Habeeb Olufowobi**, “Towards Enhanced Sparse-View Tomographic Reconstruction Using 3D Gaussian Splatting,” *International Conference on 3D Vision*, 2026.

- C23. **ESANN'26** Paul Agbaje, Arkajyoti Mitra, Afia Anjum, Pranali Khose, Ebelechukwu Nwafor, and **Habeeb Olufowobi**, "Privacy-preserving Intrusion Detection System for Internet of Vehicles Using Split Learning," *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*, 2026.
- C22. **BHI'25** Aqsa Yousaf, Paul Agbaje, Megan Coffee, and **Habeeb Olufowobi**, "Beyond Detection: Comparative Explainability Study on Trypanosoma cruzi Using CAMs and DETR Attention," *IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 1–7, 2025.
- C21. **ECAI'25** Arkajyoti Mitra, Afia Anjum, Paul Agbaje, Mert D. Pesé, and **Habeeb Olufowobi**, "FedVLM: Scalable Personalized Vision-Language Models through Federated Learning," *European Conference on Artificial Intelligence (ECAI)*, pp. 4249–4256, 2025.
- C20. **CNS'25** Afia Anjum, Arkajyoti Mitra, Paul Agbaje, Md. Ahanaful Alam, Debashri Roy, Md Salik Parwez, and **Habeeb Olufowobi**, "SemPerGe: Unveiling Text-based Adversarial Attacks on Semantic Communication," *IEEE Conference on Communications and Network Security (CNS)*, pp. 1–9, 2025.
- C19. **VehicleSec'25** Arkajyoti Mitra, Pedram MohajerAnsari, Afia Anjum, Paul Agbaje, Mert D. Pesé, and **Habeeb Olufowobi**, "Beyond the Glow: Understanding Luminescent Marker Behavior Against Autonomous Vehicle Perception Systems," *USENIX Symposium on Vehicle Security and Privacy (VehicleSec)*, pp. 195–210, 2025.
- C18. **DSN'25** Mert D. Pesé, Bulut Gozubuyuk, Eric Andrechek, **Habeeb Olufowobi**, Mohammad Hamad, and Kang G. Shin, "MichiCAN: Spoofing and Denial-of-Service Protection using Integrated CAN Controllers," *IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, pp. 443–456, 2025.
- C17. **CCNC'25** Raju Hazari, Gaurav Singh, Devika Renjith, Divya Krishnan, **Habeeb Olufowobi**, and Debashri Roy, "Spec-SCAN: Spectrum Learning in Shared Channel using Neural Networks," *IEEE Consumer Communications & Networking Conference (CCNC)*, pp. 1–6, 2025.
- C16. **GLOBECOM'24** Ebelechukwu Nwafor, Ujjwal Baskota, Md Salik Parwez, Jeremy Blackstone, and **Habeeb Olufowobi**, "Evaluating Large Language Models for Enhanced Intrusion Detection in Internet of Things Networks," *IEEE Global Communications Conference (GLOBECOM)*, pp. 3358–3363, 2024.
- C15. **IECON'24** Uchenna Ezeobi, Sena Hounsinnou, **Habeeb Olufowobi**, Yanyan Zhuang, and Gedare Bloom, "MCFICS: Model-based Coverage-guided Fuzzing for Industrial Control System Protocol Implementations," *IEEE IECON 2024-50th Annual Conference of the IEEE Industrial Electronics Society*, pp. 1–6, 2024.
- C14. **ICIT'24** Omolade Ikumapayi, Paul Agbaje, Yanyan Zhuang, **Habeeb Olufowobi**, and Gedare Bloom, "Deadline-Based Class Assignment for Time-Sensitive Network Frame Preemption," *IEEE International Conference on Industrial Technology (ICIT)*, pp. 1–8, 2024.

- C13. **VehicleSec'24** Paul Agbaje, Abraham Mookhoek, Afia Anjum, Arkajyoti Mitra, Mert D. Pesé, and **Habeeb Olufowobi**, “AutoWatch: Learning Driver Behavior with Graphs for Auto Theft Detection and Situational Awareness,” *ISOC Symposium on Vehicle Security and Privacy (VehicleSec)*, 2024.
- C12. **BDCAT'23** Paul Agbaje, Afia Anjum, Arkajyoti Mitra, Sena Hounsinnou, Ebelechukwu Nwafor, and **Habeeb Olufowobi**, “Privacy-preserving Intrusion Detection System for Internet of Vehicles Using Split Learning,” *IEEE/ACM International Conference on Big Data Computing, Applications and Technologies*, pp. 1–8, 2023.
- C11. **ICMLA'23** Ebelechukwu Nwafor, Carter Schmidt, and **Habeeb Olufowobi**, “IoT-MGSec: Mitigating Man-in-the-Middle Attacks in IoT Networks Using Graph-Based Learning,” *International Conference on Machine Learning and Applications (ICMLA)*, 2023.
- C10. **EDGE'23** Paul Agbaje, Afia Anjum, Zahidur Talukder, Mohammad Islam, Ebelechukwu Nwafor, and **Habeeb Olufowobi**, “FedCime: An Efficient Federated Learning Approach for Clients in Mobile Edge Computing,” *IEEE International Conference on Edge Computing and Communications (EDGE)*, pp. 215–220, 2023.
- C9. **INDIN'23** Paul Agbaje, Ebelechukwu Nwafor, and **Habeeb Olufowobi**, “Deep Reinforcement Learning for Energy-Efficient Task Offloading in Cooperative Vehicular Edge Networks,” *IEEE International Conference on Industrial Informatics (INDIN)*, pp. 1–8, 2023.
- C8. **ICCE'23** Afia Anjum and **Habeeb Olufowobi**, “Towards Mitigating Blackhole Attack in NDN-Enabled IoT,” *IEEE International Conference on Consumer Electronics (ICCE)*, 2023.
- C7. **ICMLA'22** Ebelechukwu Nwafor and **Habeeb Olufowobi**, “CANBERT: A Language-based Intrusion Detection Model for In-vehicle Networks,” *IEEE International Conference on Machine Learning and Applications (ICMLA)*, pp. 294–299, 2022.
- C6. **MECO'22** Afia Anjum, Paul Agbaje, Sena Hounsinnou, and **Habeeb Olufowobi**, “In-Vehicle Network Anomaly Detection Using Extreme Gradient Boosting Machine,” *Mediterranean Conference on Embedded Computing (MECO)*, pp. 1–6, 2022.
- C5. **RTSS'21** Sena Hounsinnou, Mark Stidd, Uchenna Ezeobi, **Habeeb Olufowobi**, Mitra Nasri, and Gedare Bloom, “Vulnerability of Controller Area Network to Schedule-Based Attacks,” *IEEE Real-Time Systems Symposium (RTSS)*, pp. 495–507, 2021.
- C4. **WF-IoT'21** Ebelechukwu Nwafor, Michael Robson, and **Habeeb Olufowobi**, “Dynamic Load Sharing in Memory Constrained Devices: A Survey,” *IEEE World Forum on Internet of Things (WF-IoT)*, 2021.
- C3. **Big DATA'19** Ebelechukwu Nwafor and **Habeeb Olufowobi**, “Towards an Interactive Visualization Framework for IoT Device Data Flow,” *In 2019 IEEE International Conference on Big Data (Big Data)*, IEEE, 2019.
- C2. **UEMCON'18** Md Salik Parwez and **Habeeb Olufowobi**, “Cost-constrained Handoff in Next Generation Heterogeneous Wireless Networks,” *IEEE Ubiquitous Computing, Electronics and Mobile Communication Conference (UEMCON)*, 2018.

- C1. **ICSOC'16** **Habeeb Olufowobi**, Robert Engel, Nathalie Baracaldo, Luis Angel D. Bathen, Samir Tata, and Heiko Ludwig, “Data Provenance Model for Internet of Things Systems,” *International Conference on Service-Oriented Computing*, Springer, 2016.

✔ Refereed Workshops Articles

- W9. **ICCV-W'25** Aqsa Yousaf, Arkajyoti Mitra, Paul Agbaje, Afia Anjum, and **Habeeb Olufowobi**, “DAPS-AGF: Depth-Aware Perceptual Similarity with Adaptive Gradient Filtering for Enhanced Outdoor Scene Reconstruction,” *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 2648–2656, 2025.
- W8. **RTAS-BP'23** Afia Anjum, Sena Hounsinnou, and **Habeeb Olufowobi**, “Work-in-Progress: Deadline-Aware Named Data Networking for Time-Sensitive IoT Applications,” *IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*, pp. 353–356, 2023.
- W7. **RTAS-BP'23** Omolade Ikumapayi, **Habeeb Olufowobi**, Jeremy Daily, Tingting Hu, Ivan Cibrario Bertolotti, and Gedare Bloom, “Work in Progress: Schedulability Analysis of CAN and CAN FD Authentication,” *IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*, pp. 349–352, 2023.
- W6. **CPSIoTSec'22** Vijay Banerjee, Sena Hounsinnou, **Habeeb Olufowobi**, Monowar Hasan, and Gedare Bloom, “Secure Reboots for Real-Time Cyber-Physical Systems,” *Workshop on CPS & IoT Security and Privacy*, pp. 27–33, 2022.
- W5. **AutoSec'22** Paul Agbaje, Afia Anjum, Arkajyoti Mitra, Gedare Bloom, and **Habeeb Olufowobi**, “A Framework for Consistent and Repeatable Controller Area Network IDS Evaluation,” *International Workshop on Automotive and Autonomous Vehicle Security (AutoSec)*, 2022. 🏆 **Best Paper Award**
- W4. **CPS-SPC'19** **Habeeb Olufowobi**, Sena Hounsinnou, and Gedare Bloom, “Controller Area Network Intrusion Prevention System Leveraging Fault Recovery,” *ACM Workshop on Cyber-Physical Systems Security & Privacy*, 2019.
- W3. **AutoSec'19** **Habeeb Olufowobi**, Uchenna Ezeobi, Eric Muhati, Clinton Young, Joseph Zambreno, and Gedare Bloom, “Anomaly Detection Approach Using Adaptive Cumulative Sum Algorithm for Controller Area Network,” *ACM Workshop on Automotive Cybersecurity*, 2019.
- W2. **AutoSec'19** Clinton Young, **Habeeb Olufowobi**, Joseph Zambreno, and Gedare Bloom, “Automotive Intrusion Detection Based on Constant CAN Message Frequencies Across Vehicle Driving Modes,” *ACM Workshop on Automotive Cybersecurity*, 2019.
- W1. **RTSS-WiP'18** **Habeeb Olufowobi**, Clinton Young, Joseph Zambreno, and Gedare Bloom, “WiP: Real-Time Modeling for Intrusion Detection in Automotive Controller Area Network,” *IEEE Real-Time Systems Symposium (RTSS)*, 2018.

Book Chapter

- B1. **BOOK'19** **Habeeb Olufowobi** and Gedare Bloom, “Connected Cars: Automotive Cybersecurity and Privacy for Smart Cities,” in *Smart Cities Cybersecurity and Privacy*, pp. 227–240, Elsevier, 2019.

Posters

- P3. **AIM-AHEAD'25** Aqsa Yousaf, Megan Coffee, and **Habeeb Olufowobi**, “Poster: Beyond Detection: A Trustworthy and Explainable AI Framework for Chagas Disease Using YOLOv8 and DINO-DETR,” *AIM-AHEAD Annual Meeting*, 2025.
- P2. **AIM-AHEAD'24** **Habeeb Olufowobi** and Megan Coffee, “Poster: Improving Health Equity in Algorithmic Decision-Making Using Explainable AI Techniques,” *AIM-AHEAD Annual Meeting*, 2024.
- P1. **S&P'22** Afia Anjum and **Habeeb Olufowobi**, “Poster: Mitigating Blackhole Attack in NDN-T,” *IEEE Symposium on Security and Privacy*, 2022.
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Teaching Experience

- **The University of Texas at Arlington**
 - **Fall 2024, Spring 2025**
CSE 4342/5342/6351 — Embedded Systems II
 - **Spring 2022, 2023, 2026**
CSE 5388/6388 — Advance Topics in Information Security
New course developed
 - **Spring 2024**
CSE 4380 — Information Security I
 - **Spring 2021**
CSE 4344/5344 — Computer Networks
 - **Fall 2020, 2021, 2022, 2023, 2024, 2025**
CSE 5333/4392 — Cloud Computing
New course developed
- **UNCF HBCU CS Academy**
 - **Summer 2020**
Data Structures and Algorithms
- **Howard University, Washington, DC**

- **Spring 2020**
CSCI 136 — Computer Science II: Intro to Data Structures & Algorithms
CSCI 354 — Computer Science III: Data Structures & Algorithms
 - **Fall 2019**
CSCI 365 — Cloud Computing
CSCI 354 — Computer Science III: Data Structures & Algorithms
 - **Spring 2018**
EECE 156 — Math I Laboratory
 - **Fall 2017** *Teaching Assistant*
Operating Systems Concepts
Computer Organization I & II
 - **Spring 2016** *Teaching Assistant*
Computer Organization I & II
 - **Fall 2016** *Teaching Assistant*
Modeling and Simulation
- **California State Polytechnic University, Pomona**
- **Spring 2014** *Teaching Assistant*
Electromagnetic Fields

UTA Teaching Evaluation Summary

Course	Semester	Enrollment	Evaluation
CSE 5388/6388 Advanced Topics in Information Security	Spring 2026	16	4.96
CSE 5333/4392 Cloud Computing	Fall 2025	70	4.58
CSE 4342/5342 Embedded Systems II	Spring 2025	23/5	2.94/4.40
CSE 4342 Embedded Systems II	Fall 2024	33	4.17
CSE 5333 Cloud Computing	Fall 2024	17	4.46
CSE 5333 Cloud Computing	Spring 2024	21	4.89
CSE 4380 Information Security	Spring 2024	9	4.10
CSE 4392 Special Topics in Cloud Computing	Fall 2023	55	4.10
CSE 5388/6388 Advanced Topics in Information Security	Spring 2023	7	4.71
CSE 5333/4392 Cloud Computing	Fall 2022	104/7	3.86/3.76
CSE 5388/6388 Advanced Topics in Information Security	Spring 2022	20/20	4.18/4.52
CSE 5333/4392 Cloud Computing	Fall 2021	47/7	4.28/3.88
CSE 4344/5344 Computer Networks	Spring 2021	13	4.24
CSE 5333/4392 Cloud Computing	Fall 2020	38	3.88

Average scores are on a 1–5 scale (1: strongly disagree, 5: strongly agree) for the instructor items of the SFS.

Mentoring and Supervising

Current Students

- **Ph.D. Paul Agbaje** Fall 2021 – Present
Dissertation Topic: Toward Secure and Efficient Graph Neural Networks: From Adversarial Risks to Distillation Techniques
Major Accomplishments:
 - Outstanding Doctoral Dissertation Award, College of Engineering, UTA, 2026
 - Summer 2026 Dissertation Fellowship, College of Engineering, UTA, 2026
 - Kelcy Warren Graduate Fellowship for Engineering, College of Engineering, UTA, 2024
 - Travel Grant Award, Network and Distributed System Security Symposium (NDSS), 2024
 - Participant Support Stipend, STARS Celebration, 2023
 - Tapia Registration Scholarship, ACM Richard Tapia Celebration of Diversity in Computing Conference, 2023
 - NSF Student Travel Award, Cyber-Physical Systems and Internet-of-Things Week (CPS-IoT Week), 2023
 - Outstanding Teaching Assistant, Department of Computer Science and Engineering, UTA, 2023
 -  Best Paper Award, 4th International Workshop on Automotive and Autonomous Vehicle Security (AutoSec), 2022
 - Honorable Mention, Ph.D. Lightning Talk Award, Student Computing Research Festival (SCRF), UTA, 2022
- **Ph.D. Aqsa Yousaf** Fall 2023 – Present
Dissertation Topic: Reliable AI-Assisted Disease Diagnosis in Medical Imaging
Major Accomplishments:
 - NSF–EMBS–Google Sponsored Young Professional NextGen Scholar Recognition, IEEE International Conference on Biomedical and Health Informatics (BHI), 2025.
 - Recipient of UTA Graduate School Travel Grant to present research at a BHI conference, 2025.
- **Ph.D. Joyce Odeh** Fall 2025 – Present
Research Focus: Protocol Fuzzing and Security Analysis for Cyber-Physical and Vehicular Systems

- **Ph.D. Siddhant Shettiwar** Spring 2026 – Present
Research Focus: Formal Methods and Security Analysis for Cyber-Physical and Vehicular Systems
Major Accomplishments:
 - Selected participant, SRI Summer School on Formal Techniques, May 2026
 - **Ph.D. Muhammad Samee Sevas** Fall 2025 – Present
Research Focus: Autonomous Vehicle Security
 - **Ph.D. Azaz Mir** Fall 2025 – Present
Research Focus: AI/ML Systems Security
 - **Ph.D. Myriam Ladhari** Spring 2026 – Present
Research Focus: Autonomous Vehicle Perception Systems Security
 - **M.S. Farhan Sadiq** Fall 2025 – Present
Research Focus: Agentic AI, Formal Verification, Methodologies for Secure Software Framework Upgrades and Vulnerability Detection
 - **M.S. Annjoy Njoroge** Fall 2025 – Present
Research Interests: Formal Verification and Security Analysis for Cyber-Physical and Vehicular Systems
 - **M.S. Kopil Sharma** Fall 2024 – Present
Research Interests: Adversarial Analysis of Automotive Perception Systems
 - **Undergrad Md. Ahanaful Alam** Fall 2024 – Present
Undergraduate Student, Computer Science
 - **Undergrad Abdullateef Ajala** Fall 2025 – Present
Undergraduate Student, Computer Science
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✓ Graduated Students and Alumni

- **Ph.D. Afia Anjum** Fall 2021 – Spring 2026
Current Position: Assistant Professor, Texas Christian University (TCU)
Dissertation Topic: Optimizing the Frontier: Enhancing Security and Efficiency of Wireless Communication Systems for Internet of Things
Major Accomplishments:
 - Rising Star Award, IEEE TCCN Special Interest Group on AI/ML in Security, 2025
 - Travel Grant Award, USENIX VehicleSec, 2025

- Tuition Assistance Award, Los Alamos National Laboratory: providing full support for Ph.D. tuition (2025 - 2026)
 - 🏆 Best Paper Runner-up Award, 23rd International Conference on Machine Learning and Applications, 2024
 - Invited speaker at Named Data Networking Community Meeting, 2024
 - Travel Grant Award, Network and Distributed System Security Symposium (NDSS), 2024
 - 🏆 Best Ph.D. Lightning Talk Award, Student Computing Research Festival (SCRF), UTA, 2024
 - NSF Student Travel Award, Cyber-Physical Systems and Internet-of-Things Week (CPS-IoT Week), 2023
 - 2nd Runner-up Award, Ph.D. Lightning Talk, Student Computing Research Festival (SCRF), UTA, 2023
 - Best Paper Award, 4th International Workshop on Automotive and Autonomous Vehicle Security (AutoSec), 2022
 - Student Travel Award, IEEE Symposium on Security and Privacy (IEEE S&P), 2022
 - Runners-up Award, Ph.D. Lightning Talk, Student Computing Research Festival (SCRF), UTA, 2022
- **Ph.D. Arkajyoti Mitra** Fall 2021 – Fall 2025
Current Position: Post Doctoral Fellow, UT Southwestern Medical Center (UTSW)
Dissertation Topic: Scalable, Secure, and Adaptable Perception Systems Through Adversarial Analysis and Federated Fine-Tuning
Major Accomplishments:
 - Travel Grant Award, USENIX Symposium on Vehicle Security and Privacy (VehicleSec), 2025
 - Travel and Registration Grant Award, PyTorch Conference, 2025
 - Summer 2025 Research Fellowship, College of Engineering, UTA, 2025
 - Travel Grant Award, Network and Distributed System Security Symposium (NDSS), 2024
 - Tapia Registration Scholarship, ACM Richard Tapia Celebration of Diversity in Computing Conference, 2023
 - Travel Grant Award, CyberTruck Challenge Workshop, 2022
 - Travel Grant Award, CyberTractor Challenge Workshop, 2022
 - Best Paper Award, 4th International Workshop on Automotive and Autonomous Vehicle Security (AutoSec), 2022
 - **M.S. Thesis Jesse Kiprotich Chumo** Fall 2024 – Spring 2026
Dissertation Topic: Evaluating the Robustness of GNN-Based Vulnerability Detectors Under Semantics-Preserving Code Obfuscation
 - **M.S. Thesis Mohammad Sufyaan Saeed** Spring 2024 – Fall 2025
Dissertation Topic: Training Data Privacy in Machine Learning: A Systematization of Attacks and Defenses

- **M.S. Pranali Sanjay Khose** Fall 2022 – Fall 2024
Current Position: Solutions Architect, Amazon AWS
Project Topic: Enhancing Graph Neural Networks: A Mutual Learning Approach
 - **M.S. Mayank Vekariya** Spring 2023 – Fall 2023
Current Position: Software Development Engineer, Amazon AWS
 - **Undergrad Abraham Mookhoek** Fall 2022 – Fall 2024
Current Position: Ph.D. Student at UTA
 - **Undergrad Mujaddad Fazeel** Spring 2024 – Spring 2025
 - **Undergrad Kazi Md Waliul Azim** Fall 2023 – Spring 2025
 - **Undergrad David Oyekola** Fall 2022 – Spring 2025
 - **Undergrad Turza Saha** Fall 2022 – Spring 2024
 - **Undergrad Mustafa Daher** Fall 2022 – Spring 2025
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✓ Graduate Dissertation and Thesis Committees

- **Ph.D. Dissertation Defense**
 - Dervis Ozay (External Member; Advisor: Prof. Shouyi Wang, UTA), May 2026
 - Aref Shiran (Committee Member; Advisor: Prof. Younghe Liu, UTA), May 2026
 - Trivedi Bhaskar (Committee Member; Advisor: Prof. Manfred Huber, UTA), December 2026
 - Mohit Singhal (Committee Member; Advisor: Prof. Shirin Nilizadeh, UTA), May 2024
 - Uchenna Ezeobi (External Member; Advisor: Prof. Gedare Bloom, UCCS), July 2024
- **M.S. Thesis Defense**
 - Uzma Zehra (Committee Member; Advisor: Prof. Md Salik Parwez, UTA), May 2026
 - Amnah Abdelrahman (Committee Member; Advisor: Prof. Yingying Zhu, UTA), May 2023
- **Ph.D. Dissertation Committee**
 - Tennov Simanjuntak (Committee Member; Advisor: Prof. Christoph Csallner, UTA) Spring 2024 – Present
 - Shaikh Islam (Committee Member; Advisor: Prof. Dianqi Han, UTA) Fall 2025 – Present

Honors and Awards

Research and Scholarly Awards

- Outstanding Early Career Research Award April 2026
College of Engineering, University of Texas at Arlington
- Blossoms of Discovery – Tenure-Track Research Award November 2025
Department of Computer Science and Engineering, University of Texas at Arlington
- Ronald E. McNair Scholars Program & Louis Stokes Alliance for Minority Participation Mentors Award April 2025
University of Texas at Arlington
- Rising Star Teaching Award November 2024
Department of Computer Science and Engineering, University of Texas at Arlington
- Google-CAHSI Cybersecurity Curriculum Fellowship Award October 2024
Computing Alliance of Hispanic-Serving Institutions (CAHSI)
- Distinguished Alumnus Award January 2024
Fountain University, Osogbo, Nigeria
- AIM-AHEAD Research Fellowship Award September 2023
The National Institutes of Health’s AIM-AHEAD program
- Best Paper Award, May 2022
International Workshop on Automotive and Autonomous Vehicle Security (AutoSec)’22@NDSS
- HBCU Blockchain Course Development Proposal Award October 2019
Morgan State University
- Dean’s List 2014
California State Polytechnic University, Pomona
- Chancellor’s Prize – Best Graduating Student in the University July 2011
Fountain University, Osogbo, Nigeria
- College Prize – Best Graduating Student in Sciences July 2011
Fountain University, Osogbo, Nigeria
- Fountain University Scholar Award 2008 – 2011
Fountain University, Osogbo, Nigeria

Travel Awards

- Real-Time System Symposium Travel Award April 2018
NSF-Funded Travel Support
- Trans-Atlantic Symposium on ICT Technology and Policy Travel Award June 2017
NSF-Funded Travel Support
- CPS Week Student Travel Award April 2017
Carnegie Mellon University

- IEEE Security Development Conference Travel Award
IEEE SecDev Conference

October 2016

⊗ Professional Service

✔ Conference Organization

- **TPC Chair** – 4th EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP) 2026
- **Poster and Demo Chair** – IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM) 2025
- **Web Chair** – Symposium on Vehicle Security and Privacy (VehicleSec) 2024
- Artifacts Evaluation Committee Member – Real Time System Symposium (RTSS) 2021, 2022, 2023

✔ Technical Program Committee

- International Joint Conference on Artificial Intelligence (IJCAI), 2025, 2026
- USENIX Symposium on Vehicle Security and Privacy (VehicleSec), 2026
- IEEE Intelligent Vehicle Symposium (IV), 2026
- International Conference on Computer Communications and Networks (ICCCN) 2026
- Silicon Valley Cybersecurity Conference, 2026
- IEEE International Conference on Communications (ICC) 2024, 2025, 2026
- Real Time System Symposium (RTSS) 2021, 2022, 2023, 2025
- The ACM Web Conference (WWW) 2026
- ACM International Conference on Multimedia (ACMMM) 2026
- Annual AAAI Conference on Artificial Intelligence SAPP 2026
- IEEE International Conference on Embedded and Real-Time Computing Systems and Application (RTCSA) 2024
- IEEE/ACM International Conference on Utility and Cloud Computing (UCC) 2022, 2023

✔ Journal and Conference Reviewing

- IEEE Transactions on Vehicular Technology
- IEEE Internet of Things Journal
- IEEE Transactions on Intelligent Transportation Systems
- Elsevier Computers & Security

- Future Generation Computer Systems
- Computer Networks
- The British Machine Vision Conference (BMVC) 2025
- ACM SIGCSE

✓ Federal Agency Review Panels

- National Science Foundation 2023

🏢 Institutional and Departmental Service

- Faculty Advisor: National Society of Black Engineers (NSBE) Fall 2022 - Present
College of Engineering, University of Texas at Arlington
- Member, CAE-R Cybersecurity Designation Committee Spring 2024 – Present
College of Engineering, University of Texas at Arlington
- Member, Faculty and Staff Awards Committee Spring 2025 – Present
Computer Science and Engineering Department, University of Texas at Arlington
- Cybersecurity Degree Planning Committee Spring 2025 – Present
Computer Science and Engineering Department, University of Texas at Arlington
- Member, International Program Spring 2025 – Present
Computer Science and Engineering Department, University of Texas at Arlington
- GTA Orientation Committee Fall 2024 – Spring 2025
Computer Science and Engineering Department, University of Texas at Arlington
- Member, Broadening Participation Committee Spring 2022 – 2023
Computer Science and Engineering Department, University of Texas at Arlington
- CSE T/TT faculty search committee Fall 2022 – Spring 2023 Computer Science and
Engineering Department, University of Texas at Arlington
- Member, PhD Admissions Committee Fall 2020 – Spring 2023
Computer Science and Engineering Department, University of Texas at Arlington

🎤 Invited Talks

- *Current Trends in Data Science: Innovations, Challenges, and Future Directions* March 2025
Howard University's Research Center in Minority Institutions (RCMI) Program's - Virtual
Applied Data Science Training Institute (VADSTI)
- *Explainable AI and Interpretability in Algorithm Decision Making* March 2024
Howard University's Research Center in Minority Institutions (RCMI) Program's - Virtual
Applied Data Science Training Institute (VADSTI)

- *Algorithmic Bias and Real-World Consequences* February 2024
Howard University's Research Center in Minority Institutions (RCMI) Program's - Virtual Applied Data Science Training Institute (VADSTI)
 - *Cloud Computing II* March 2023
Howard University's Research Center in Minority Institutions (RCMI) Program's - Virtual Applied Data Science Training Institute (VADSTI)
 - *Intrusion Detection and Prevention System for Ground Vehicles* October 2022
Toyota Connected, Dallas
 - *Remote Compromise of Automotive Communication Network* March 2022
ISACA North Texas Annual Meeting
 - *Securing Vehicle-to-Vehicle Communications* August 2021
Fountain University, Osogbo Nigeria
 - *Security of Ground and Urban Air Mobility Communication* September 2020
Bell Flight, Fort Worth
 - *Autonomous Vehicle Security* April 2020
Florida A&M University Thought Leaders Series
 - *Automotive Intrusion Detection using CAN Timing* May 2019
University of Nevada Las Vegas
 - *Provenance Model in IoT System* March 2019
Fairleigh Dickinson University Teaneck
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Presentations

- *FedVLM: Scalable Personalized Vision-Language Models through Federated Learning*
ECAI 2025
- *SemPerGe: Unveiling Text-based Adversarial Attacks on Semantic Communication*
CNS 2025
- *Beyond Detection: A Trustworthy and Explainable AI Framework for Chagas Disease Using YOLOv8 and DINO-DETR*
AIHES 2025
- *Current Trends in Data Science: Innovations, Challenges, and Future Directions*
VADSTI 2025
- *Improving Health Equity in Algorithmic Decision-Making Using Explainable AI*
AIHES 2024
- *Algorithmic Bias and its real-World Consequences*
VADSTI 2024
- *XAI and Interpretability in Algorithm Decision Making - Understanding the Black Box*
VADSTI 2024

- *Towards Mitigating Blackhole Attack in NDN-Enabled IoT*
ICCE 2023
 - *Automotive Intrusion Detection Based on Constant CAN Message Frequencies Across Vehicle Driving Modes*
AutoSec 2019
 - *Anomaly Detection Approach Using Adaptive Cumulative Sum Algorithm for Controller Area Network*
AutoSec 2019
 - *Cost-constrained Handoff in Next Generation Heterogeneous Wireless Networks*
UEMCON 2018
 - *Real-Time Modeling for Intrusion Detection in Automotive Controller Area Network*
RTSS 2018
 - *Data Provenance Model for Internet of Things Systems*
IBM Almaden Research Center, Almaden 2016
 - *Secure Data Provenance for Internet of Things Devices (IoT)*
HU Research Week 2016
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Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE) Senior Member
- IEEE Computer Society
- IEEE Communication Society